Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829
Tax ID 0575566002051

CASE STUDY · CUSTOMER QUESTION

Solar report vs PEA bill

"Why does my PEA bill show more units than my inverter app?" · 5.08 kWp hybrid with battery · Pathum Thani · Apr–Oct 2026

6,466 kWh Solar produced Apr–Oct

4,458 units PEA billed Apr–Oct

1,765 kWh Plant recorded from grid

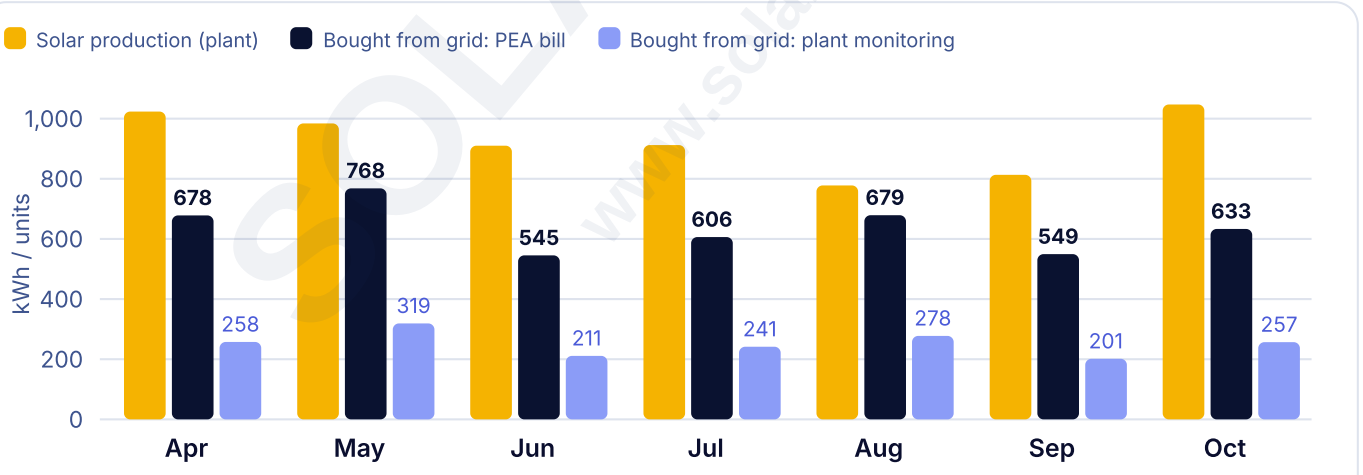
2.5× PEA vs plant grid reading
+2,693 units gap

1 October 2026 at a glance

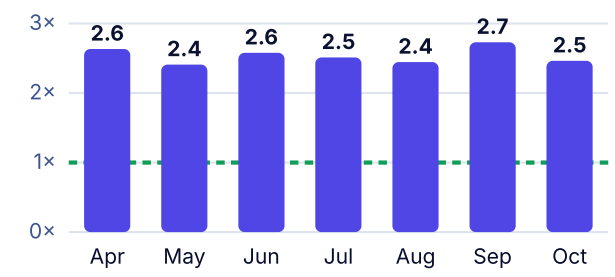
	PEA meter (October bill)	Solar plant monitoring	Gap
Bought from grid	633 units	257 kWh	2.5×
Sent to grid	147 units	62.5 kWh	2.4×
Solar production	–	1,046.4 kWh	
Battery charged / discharged	–	435.9 / 383.6 kWh	

October bill: 633 units (Peak 70 · Off-peak 563). PEA also bought back the 147 units the meter recorded as sent to the grid.

2 Month by month: solar production and grid units

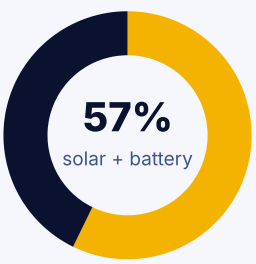


3 PEA ÷ plant, every month



- - - 1× = the two readings should match

4 Where October's energy really came from



847 kWh solar + battery
633 units from PEA grid
Real use ≈ **1,480 kWh**
(app shows 1,189 kWh)

SOLAR ESS www.solaress.org

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5 The numbers, month by month

Month 2026	Solar production (kWh)	Plant: bought from grid (kWh)	PEA bill (units)	Difference	PEA ÷ plant
Apr	1,023.5	257.5	**678**	+420.5	**2.63×**
May	983.7	318.9	**768**	+449.1	**2.41×**
Jun	909.8	211.3	**545**	+333.7	**2.58×**
Jul	912.0	241.2	**606**	+364.8	**2.51×**
Aug	777.6	277.7	**679**	+401.3	**2.45×**
Sep	812.8	201.0	**549**	+348.0	**2.73×**
Oct	1,046.4	257.0	**633**	+376.0	**2.46×**
Total	**6,465.8**	**1,764.6**	**4,458**	**+2,693.4**	**2.53×**

Each PEA bill is read on the 1st of the following month (bill 10/2569 read 01/11/2569), so it covers the same calendar month as the plant report.

6 What this means

The solar plant is producing well

About 780–1,050 kWh a month, 6.47 MWh from April to October. That is normal for a 5.08 kWp system.

The grid reading is too low, both ways

PEA counted 2.5× more units bought and 2.4× more units sold than the plant recorded. A steady ratio in both directions points to the CT (grid current sensor), not to loads bypassing it.

Real home use is higher than the app shows

With PEA's figures, October use was about 1,480 kWh, not 1,189. Solar and battery covered about 57%; the grid supplied the rest.

Battery and export control are affected

The inverter decides when to charge, discharge and export from the CT reading. Seeing only about 40% of the grid import, it can leave the battery idle while the home still buys power.

Recommended next step: CT and meter check on site

- Find the CT: on the main incoming cable after the PEA meter, with every house circuit downstream of it.
- Check the CT clamps around the whole conductor (all parallel cables, one phase only) and the arrow points the right way.
- Check the CT / meter ratio setting in the inverter matches the CT fitted.
- Compare a clamp-meter reading with the inverter's grid power and the PEA meter at the same moment.
- Re-check after one week: PEA units and plant import should match within a few percent.

Sources: the customer's PEA bill 10/2569 and PEA Smart Plus usage history; plant monitoring export "PlantsDetails-History" (monthly totals, March–November 2026). October real use = solar 1,046.4 + grid 633 − export 147 + battery discharge 383.6 − charge 435.9.